

Week 6, Day 1 — student handout

Umesh Indrajith · Refactor into handlers. No new API features required.

OUTCOMES

- Function parts; multiple returns; do not ignore err.
- defer; methods with pointer receivers; New... constructors.
- Move health/hello into internal/handlers.
- FullName + green tests.

HANDLER SHAPE

```
package handlers

type HealthHandler struct{}

func NewHealthHandler() *HealthHandler {
    return &HealthHandler{}
}

func (h *HealthHandler) Get(c *gin.Context) {
    c.JSON(http.StatusOK, gin.H{
        "status": "ok",
        "message": "Backend is running",
    })
}
```

THIN MAIN

```
health := handlers.NewHealthHandler()
hello := handlers.NewHelloHandler()
r.GET("/health", health.Get)
r.GET("/api/hello", hello.Get)
r.Run(":8080")
```

FULLNAME + TEST

```
go test ./internal/handlers -v
```

WORDS

Term	Meaning
Method	Function with a receiver
defer	Runs at function exit

IF IT BREAKS

- 404 → register h.Get not h.Get()
- nil panic → call New... first
- import fail → check module path
- no tests → run from backend/

NewX	Constructor function
*	Pointer receiver (preview)

HOMEWORK

PR the refactor. Health must stay green. Details: `homework.md`.